

Check fitted flocker model

Specify model to check:

```
spec <- params$spec #final_species_eco_sorted[6] # 1
spec
```

```
[1] "Downy Woodpecker"
```

```
buffer_km <- params$buffer_km
buffer_km
```

```
[1] 750
```

```
fold <- params$fold
# fold

results_dir <- params$results_dir
results_dir
```

```
[1] "M:/Documents/DEBTs/analysis/Schifferle_BBS_occupancy_models_2023/results/full_model"
```

Load fitted model:

```
load(file = file.path(results_dir,
                       paste0("out_", spec, "_fm_buffer", buffer_km, ".RData"))) # xx adjust
#print(out)

posterior <- brms::as_draws(out)
```

Check whether MCMC worked:

Do we have divergent transitions?

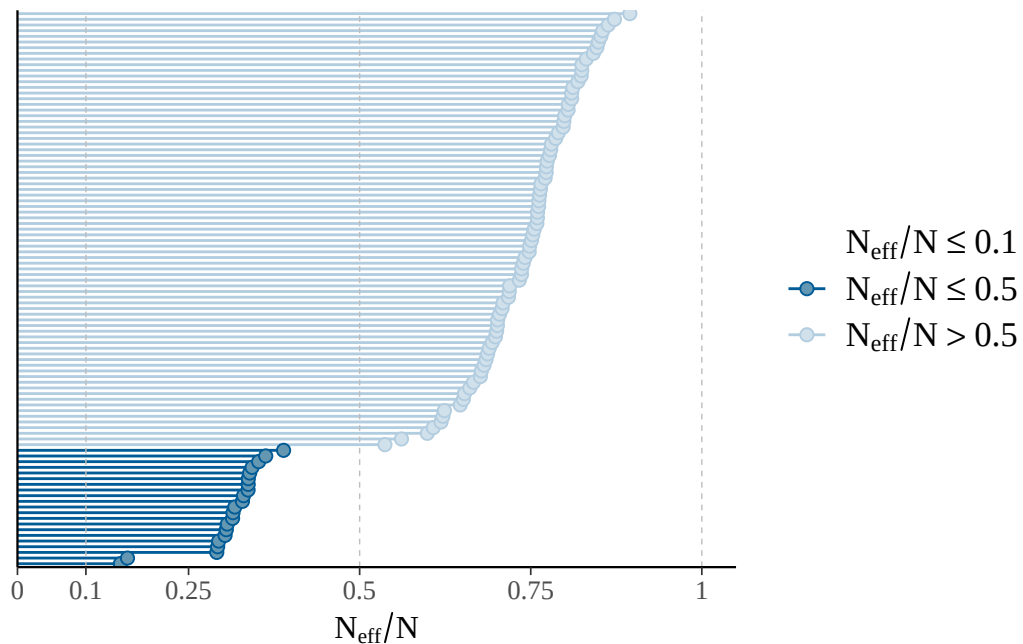
```
[1] "divergent transitions: 0"
```

Explorations regarding divergent transitions:

...

Is effective number of MCMC samples large enough:

The larger the ratio of n_{eff} to N (number of samples), the better.



```
[1] "effective sample size fine"
```

```
[1] "n_eff ratio bulk ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1836	0.6447	0.8723	0.8785	1.0913	2.1321

```
[1] "n_eff ratio tail ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1504	0.6527	0.7480	0.7036	0.7874	0.8948

```
[1] "effective sample size in bulk and tail fine"
```

Do chains mix well:

No divergences to plot.

No divergences to plot.



No divergences to plot.



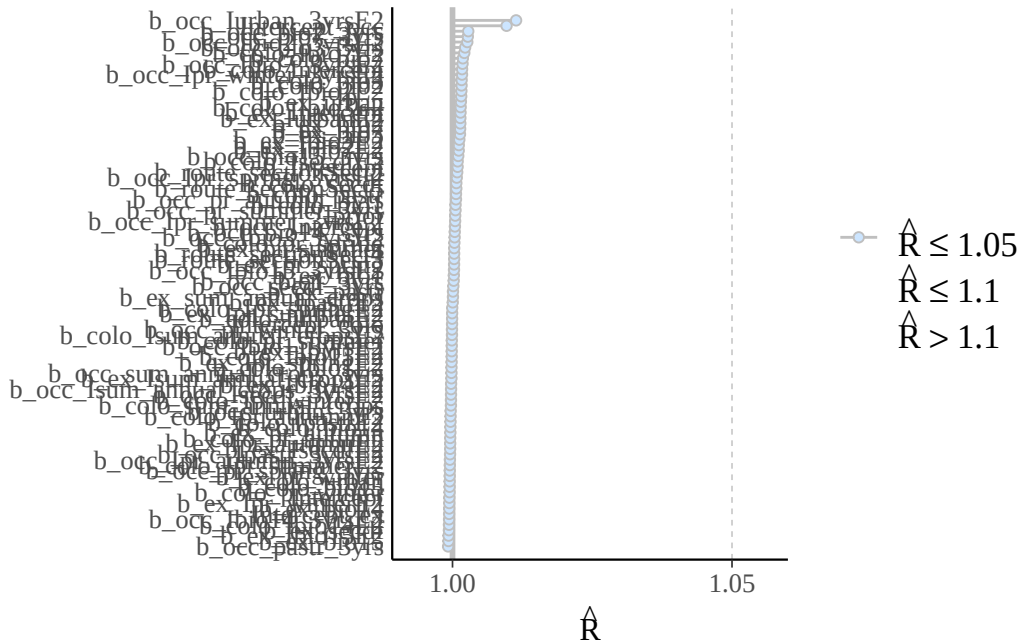
Post-warmup iteration



Post-warmup iteration

Are R-hat values fine:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9992	0.9997	1.0002	1.0006	1.0011	1.0114



Print model posteriors

```
[1] "parameters for which uncertainty interval doesn't overlap zero:"
```

```
[1] "b_occ_Intercept"      "b_occ_Ipr_spring_3yrsE2"
```

```
[1] "b_colo_bio1"      "b_colo_pr_summer" "b_colo_lbio1E2"   "b_colo_secdf"
```

```
[5] "b_colo_pastr"      "b_colo_IsecdfE2"  "b_colo_IpastrE2"
```

```
[1] "b_ex_Intercept"      "b_ex_bio1"           "b_ex_bio2"
[4] "b_ex_bio3"           "b_ex_bio7"           "b_ex_pr_summer"
[7] "b_ex_Ibio1E2"        "b_ex_Ibio2E2"        "b_ex_Ibio3E2"
[10] "b_ex_Ibio7E2"        "b_ex_Ipr_summerE2"   "b_ex_secdf"
[13] "b_ex_IurbanE2"
```

